Calculus 1m - 104018 - Spring Semester

Final Exam B

Instructions:

You must answer on these pages. You may not add pages or papers to this exam.
You must write with a black or a blue pen. Do not use a pencil or a red pen.

You may not use notes, books, calculators or electronic devices of any kind.

Good Luck!

Part A

Mark the correct answer in each of the questions. Each of the first 4 questions is 8 points.

1. Let $f(x) = \begin{cases} e^{\frac{1}{x(x-1)}} & x \neq 0,1 \\ 0 & x = 0 \\ 1 & x = 1 \end{cases}$

   Mark the correct answer.

   (a) $f(x)$ is continuous at 1.

   (b) The function is one to one (injective) on $(0, \infty)$.

   (c) $\int\limits_{2}^{\infty} f(x)dx < \infty$.

   (d) $f(x)$ is increasing on $(0, \infty)$.

   (e) $\int\limits_{0}^{1} f(x)dx < 1$.

2. Consider the improper integrals

$$I = \int_{1}^{\infty} 1 - e^{(x^{-1})} dx \quad II = \int_{0}^{1} \frac{1}{2 - 2\cos\left(\sqrt{x}\right)} dx.$$

   (a) Both improper integrals converge.

   (b) None of the improper integrals converges.

   (c) I does not converge and II converges.

   (d) I converges and II does not converge.

3. Consider the series

$$I = \sum_{n=1}^{\infty} \arctan\left(\frac{(-1)^n}{n}\right) \quad II = \sum_{n=1}^{\infty} \frac{\cos\left(\frac{1}{n}\right) + \sin\left(\frac{1}{n}\right)}{(n+1)^2 \ln\left(1 + \frac{1}{n}\right)}$$

   (a) Series I converges and series II does not converge.

   (b) Series I does not converge and series II converges.

   (c) Both series do not converge.

   (d) Both series converge.

4. Let $f(x)$ be defined for all x. Mark the correct statement.

   (a) If f is continuous at every point, and $f'(x)$ exists and is positive except at finitely many points, then f is strictly increasing.

   (b) If $f(x)$ is differentiable, then so is $f(x) + \left(\left(f(x)\right)^2\right)^{\frac{1}{4}}$

   (c) If $f(x)$ is increasing and continuous, then f is differentiable.

   (d) If f is continuous and $\lim\limits_{h\to 0} \frac{f(1+h)-f(1-h)}{h} = L$ then $f'(1)$ exists and is equal to $\frac{L}{2}$.

5. (12 points) Answer each of the following True or False questions:

(a) If a_n has a limit and $\lim\limits_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ exists, then $\lim\limits_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \leq 1$.

True False

(b) If $\sum\limits_{n=1}^{\infty} \big(2a_n + 3b_n\big)$, $\sum\limits_{n=1}^{\infty} \big(5a_n + 2b_n\big)$ converge, then $\sum\limits_{n=1}^{\infty} a_n$ converges.

True False

(c) If $\sum\limits_{n=1}^{\infty} a_n$ converges, then $\sum\limits_{n=1}^{\infty} \frac{(-1)^n \cdot a_n}{\sqrt{n}}$ converges.

True False

(d) $\lim\limits_{x\to\infty} x^{\frac{1}{x}} = 1$.

True False

Part B

Answer each of the following questions fully. Explain all your answers.

6. (28 points)

   (a) (5 points) State Lagrange's theorem.

   (b) (12 points) Prove Lagrange's theorem.

(c) (11 points) Let f be a continuous function on the interval $[0, 1)$ which is differentiable on the interval $(0, 1)$.
Prove that if $\lim_{x \to 0^+} f'(x)$ exists and is equal to L, then $f'_+(0)$ exists and is equal to L.

7.(a) (9 points) Calculate

$$\int \frac{2x+1}{x^2-4x+3}dx.$$

(b) (10 points) Calculate $\ln(1.1)$ with accuracy 10^{-3}.

(c) (9 points) Calculate

$$\lim_{x \to -1} \frac{\int_{x^2}^{x^3} \sin(t^3)dt}{\cos\left(\frac{\pi \cdot x}{2}\right)}$$